Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 N friction force opposes a 5 m displacement. Find work, including sign and units.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and record actual trials separately.

Worked reasoning: Supplied measurements give net work 12 J and kinetic-energy change 11 J. Their 1 J difference could reflect uncertainty or omitted work; it is evidence to investigate.

Your turn: Draw two bars for these measurements and label one check of the experiment.

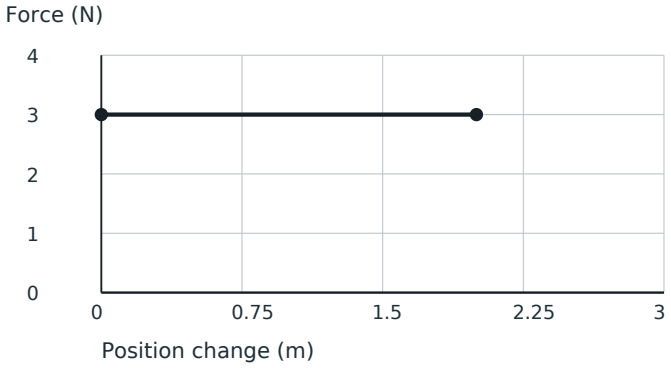
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

The supplied force-displacement graph stays at 3 N over 2 m. Calculate work.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart slows from 3 m/s to 2 m/s during a measured interval. A force sensor reports +3 J of net work over that same interval. Find initial K, final K and ΔK. Do the two measurements agree? Suggest a check.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Create a trial table with mass, initial/final speed, work and ΔK. Name two causes of disagreement that deserve checking.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two equal 4 kg carts approach at 6 m/s in opposite directions and stick. Find their common final velocity and explain.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

Why must the work and speed change refer to the same start and end positions? Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two trials have differences 1 J and 5 J, each with possible difference uncertainty ±2 J. Which is consistent with zero under those bounds? Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3